

Assignment module 3 : Understanding and Maintenance of network

Section 3: Short Answer

8. Explain the difference between a hub and a switch in a computer network.
 - **Hub:** A hub is like a loudspeaker at a party. When one computer sends data to the hub, it *shouts* it out to all the other computers connected to it. Every computer hears the message, but only the one it's meant for uses it. This can slow things down and isn't very secure.
 - **Switch:** A switch is like a smart mail sorter. When a computer sends data to a switch, the switch *knows* exactly where to send it. It delivers the message only to the computer it's meant for. This makes the network faster and more secure.
9. Describe the process of troubleshooting network connectivity issues.
 1. **Check cables and power:** Make sure everything is plugged in properly and the router or modem is turned on.
 2. **Restart devices:** Turn off your computer, router, and modem. Wait a few seconds, then turn them back on. This often fixes simple problems.
 3. **Check Wi-Fi or network connection:** See if your device is connected to the right Wi-Fi network or if the network cable is securely connected.
 4. **Look for error messages:** If your computer shows a message like "No Internet," it can give you a clue about what's wrong.
 5. **Try another device:** Check if other devices (like a phone or another computer) can connect. If they can, the problem is probably with your device.
 6. **Check IP settings:** Make sure your device is getting an IP address automatically, or check the manual settings if you're using them.
 7. **Run network troubleshooter** (on Windows/Mac): These built-in tools can often find and fix problems automatically.
 8. **Contact your Internet provider:** If nothing works, the issue might be with your internet service. Call them to check if there's an outage or other problem.

Section 4: Practical Application

10. Demonstrate how to configure a wireless router's security settings to enhance network security.

→ How to Secure Our Wireless Router

1. Connect to Our router

- Open a web browser on a computer or phone connected to your Wi-Fi.
- Type our router's IP address in the address bar. (Usually something like `192.168.0.1` or `192.168.1.1`)
- Press **Enter**.

2. Log in to the router

- Enter the **username and password**. (Default is often `admin` / `admin` or check our router's label.)
- Change the default password if we haven't already!

3. Change the Wi-Fi network name (SSID)

- Find the **Wireless Settings** or **Wi-Fi Settings** menu.
- Change the **SSID** to a unique name (but don't include personal info).

4. Set a strong Wi-Fi password

- Choose **WPA2** or **WPA3** security (these are the safest options).
- Set a **strong password** that's hard to guess (use letters, numbers, symbols).

5. Disable WPS (Wi-Fi Protected Setup)

- Go to the WPS settings and **turn it off** – it's convenient but less secure.

6. Update the router firmware

- Check for **Firmware Updates** in the settings.
- Updating keeps our router protected from new threats.

7. Disable remote management

- Unless we really need it, turn off **Remote Access** or **Remote Management**. This prevents outside access to our router settings.

8. Save settings and restart router

- Click **Save** or **Apply**.
- Restart our router to apply the changes.

Section 5: Essay

11. Discuss the importance of network documentation and provide examples of information that should be documented.

→ **Why Network Documentation is Important**

Network documentation means writing down details about how our network is set up. It helps people understand, manage, fix, and upgrade the network easily.

Why it's important:

- **Saves time** when fixing problems.
- **Helps new staff** understand the network quickly.
- **Prevents mistakes** when making changes.
- **Keeps a record** of settings and equipment used.

Examples of What to Document:

1. **Network diagram**
 - A map showing how all devices (computers, routers, switches) are connected.
2. **IP address list**
 - A table showing what IP address is given to each device.
3. **Device details**
 - Brand, model, serial number, and location of routers, switches, servers, etc.
4. **Login credentials**
 - Usernames and passwords for routers, switches, and admin tools (stored securely!).

5. **Cable layout**

- Where network cables run and what they connect.

6. **Wi-Fi settings**

- Network names (SSIDs), passwords, and encryption type (WPA2, WPA3).

7. **Backup information**

- When and how backups are made, and where they are stored.

8. **Vendor/support contacts**

- Phone numbers or emails of your internet provider or equipment supplier.